TPCNN: Two-Phase Patch-Based Convolutional Neural Network for Automatic Brain Tumor Segmentation and Survival Prediction

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Abstract. The aim of this paper is to integrate some advanced statistical methods with modern deep learning methods for tumor segmentation and survival time prediction in the BraTS 2017 challenge. The goals of the BraTS 2017 challenge are to utilize multi-institutional pre-operative MRI scans to segment out different tumor subregions and then to use tumor information to predict patient's overall survival. We build a two-phase patch-based convolutional neural network (TPCNN) model to classify all the pixels in the brain and further refine the segmentation results by using XGBoost and a post-processing procedure. The segmentation results are then used to extract various informative radiomic features for prediction of the survival time by using the XGBoost method.

Keywords: Convolutional neural network · Patch-based · XGBoost

1 Introduction

Brain region segmentation plays an important role in accurate diagnosis and efficient treatment of brain related diseases, such as stroke, Alzheimer, and glioblastoma multiforme, which has received a lot of attention in the field of computer vision, medical image analysis, and cancer imaging. Certain kinds of brain tumors such as gliomas and glioblastomas are often difficult to localize, since they are often diffused, poorly contrasted, and variously shaped without a fixed size or appearing location. Moreover, manual segmentation can be very time consuming and laborious, and therefore there is a great interest in the development of various reliable automatic brain tumor segmentation methods.

The BraTS 2017 challenge is one of such platforms, in which different segmentation methods could be extensively tested and evaluated [6,9,10]. Besides the brain tumor sub-region segmentation competition in previous years' challenges, BraTS 2017 includes an additional task of predicting patient's overall survival time by integrating radiomic features extracted from different imaging

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modalities, locations of different tumor regions, and other demographic information, such as age. A number of segmentation methods have been proposed to integrate information from different imaging modalities for automatic tumor segmentation [13]. For instance, the winner of BraTS 2013 challenge introduced a supervised segmentation framework by using multiple modality intensity, geometry, and asymmetry features, which generate the probability maps of each class by a random-forest algorithm.

With the development of deep learning methods, convolutional neural network has became more and more popular in dealing with the image segmentation problems. [7] divided the whole image of each modality into 33×33 small patches each labeled as the center annotation of the corresponding patch, and generates a single-branch CNN network with six convolutional layers, which achieves good performance on BraTS 2013 data. [3] extended the network to two-branch structure by adding another kind of patches with the same center, but larger patch size as the input, where the two branches were merged together at a certain layer before the final softmax layer. The two branch structure improves the segmentation results compared to the one-branch setting by simultaneously providing a global view and catching the location information of the corresponding pixels. [15] also adopted this two-branch framework, but they added more layers and used conditional random field, which achieve the best performance in the BraTS 2015 challenge.

Different from strategies above which predict a single pixel with a corresponding patch, fully convolutional neural network (FCNN) is proved to be a good deep learning structure in neural image segmentation since it is more efficient by dealing with a small region each time instead of the center only and thus provides a more smooth segmentation results [4]. Moreover, 2D patches are replaced by 3D patches in order to fully exploit dense-inference. However, 3D models have many more parameters compared to 2D models, which may require much longer time to train. Therefore, these kind of 'deep' 3D CNN models are not always acceptable with the limitation in computation power and time. The aim of this paper is to build a two-step strategy, in which the first step is to use a 2D model to mask out the tumor regions, and the second step is to use XGBoost to integrate the results of several simple 2D or 3D models for further refinement [1].

Although there are a lot of efforts in predicting patient's survival by using all kinds of informative radiomic features [5,14], most existing studies only use traditional quantitative features directly calculated from raw images. Little has been done on the use of abstract imaging features learned by deep learning methods for survival prediction. We integrate both quantitative radiomic features and highly abstract imaging features extracted by training a CNN model for predicting the overall survival time. Our final prediction model is built under the framework of XGBoost algorithm, which selects the most significant features, while reducing the potential risk of overfitting caused by the small sample size.

2 Data Description

The BraTS 2017 challenge data set includes subjects from three different resources, which are labeled as '2013', 'CBICA', and 'TCIA'. Specially, in the training data set, there are 20 high-grade gliomas subjects (HGGs) from group '2013', 88 HGGs from group 'CBICA', and 102 HGGs from 'TCIA', whereas 75 other subjects from group '2013' and 'TCIA' are with low-grade gliomas (LGG), which are less aggressive and infiltrative. Each patient has T2-weighted fluid attenuated inversion recovery (Flair), T1-weighted (T1), post-contrast T1-weighted (T1Gd), and T2-weighted (T2) images.

Same as the previous BraTS challenges, a validation data set is provided to test all segmentation methods and tune their associated parameters, while the final prediction on an independent test data set will be used to evaluate the finite sample performance of each participating team. Any information about the HGG/LGG status is unknown about the 46 subjects in the validation data set and the other 146 subject in the test data set.

All the four kinds of MRI scans are roughly registered to a common $240 \times 240 \times 155$ template and resampled at 1 mm resolution level. In the training data set, the annotation of three tumor regions, namely, the GD-enhancing tumor (ET, labeled 4), the peritumoral edema (ED, labeled 2), and the necrotic and non-enhancing tumor (NCR/NET, labeled 1), are provided along with the MRI scans. For the survival time prediction part, age information of 165 selected patients are also provided, which can be combined with the imaging modalities and segmented tumor regions to build and train a survival prediction model. The final predictive model is evaluated using the test data set.

3 Two-Phase Convolutional Neural Network

In this section, we discuss the details of our framework for brain tumor segmentation and how the two-phase procedures work in steps. Specifically, we use a two-branch convolutional neural network with two kinds of input patches in different sizes to briefly mask out the tumor regions, and four other 2D and 3D CNN (FCNN) models to jointly refine the segmentation within the mask determined at the first phase. Furthermore, we also discuss how we do the preprocessing to prepare the data for the deep learning models and post-processing to smooth and optimize the segmentation results.

3.1 Data Preprocessing

We find that due to the multi-institutional data origin, the raw MRI scans from different subjects suffer from varied intensity ranges. To achieve consistency in dynamic range and to ensure comparability across subjects, we use the robust intensity normalization method proposed by [15] to preprocess the images. The details of the procedures are described in the following steps:

1. Linearly transfer the original intensities of each individual MRI scans ($I_1(v)$) to the range between 0 and 255 ($I_2(v)$).
2. Calculate the intensity histogram with 256 bins. Then find the gray-value of highest histogram bin $x \in [0, 255]$ of $I_2(v)$.
3. Calculate the robust deviation which is formulated as $\sigma^2 = [\sum(v_i - x)^2]^{-1}, v_i \in I_2(v)$
4. Update the image value by $v_i^* = (v_i - x)/\sigma * c_1 + c_2$
5. Update the image value by $v_i^* = \min(\max(v_i^*, 0), 255)$

Figure 1 demonstrates comparison between before and after preprocessment for T1, FLAIR and T2 (left to right) images of a randomly selected subject.

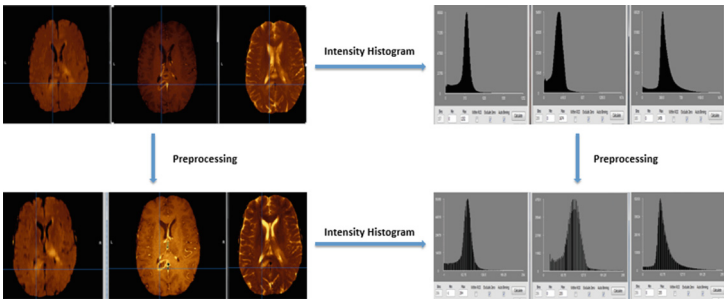

Fig. 1. Visualization of intensities before and after pre-processing

3.2 Method and Framework

After the preprocessing of the raw MRI images, we propose a patch-based two-phase convolutional neural network (TPCNN) to label all the pixels within the brain regions as non-tumor or the three tumor classes: ecrosis core, edema, enhancing core. Specifically, the second-phase could be seen as a conditional optimization procedure, which is to refine the segmentation results within the tumor mask built by the first-phase CNN model.

The input for first-phase CNN model will be the squared patches extracted from the axial slices of the MR images, where the corresponding label for each individual patch will be the label of its center pixel. In the second phase, we only need to pay attention to the regions masked by phase one. Both 3d and 2d models with different architectures and input dimensions are used, so the input patches can not only be the squares like phase one but also cubics.

Since the four modalities T1, T2, T1Gd and Flair are co-registered to a common template, for pixels at the same location of T1Gd, T2 and Flair, we generate three patches from each of them as the three channels of the input X for the CNN models, where T1 is dropped since the performance of our method with and without T1 do not show much difference and the model with fewer image channels have fewer parameters, which may drastically reduce computational time in training and testing.

In the first phase of our CNN models, we extract patches in two different sizes, where the smaller one is $(18 + 3n) \times (18 + 3n)$ and the larger one is $(35 + 6n) \times (35 + 6n)$, where n here is a kernel parameter, which determines the sizes of the two patches. The reason of this architecture choice is that we want the class prediction of a pixel to be decided by both the local features within a small region around the pixel and the larger 'context': the comparative location of the pixel inside the brain. We adopt the structure of the CNN model described in [15] in the first phase, where the larger input is merged with the small input at a certain layer after passing through several convolutional and pooling layers. The layer settings are shown in Fig. 2. By comparing the mean dice-ratios achieved by different n from experiment, we eventually choose $n = 5$ to be optimal kernel size, with which we are able to roughly mask out the whole tumor regions as the starting point for the second-phase procedure.

One major issue we find with only one-phase CNN model is that we could only briefly mask out the whole tumor region but many non-tumor pixels are misclassified as tumors. Also, the boundaries between different classes are difficult to classify, since pixels belonging to different groups along the boundaries may share similar features when the patch size is large. The high sensitivities of the first-phase results motivate us to add the second phase to refine the segmentation by resampling within the mask built by phase 1 and reduce the patch size to cover more local variations. Another advantage of using the two-phase approach is that we do not require an extremely large training sample at each individual phase which may help reduce the computation cost and save a considerable amount of running time.

We adopt four convolutional neural networks at phase two with different input sizes and try to combine the prediction results by each of them. The first model still uses the same setting as the model in phase 1, but all the sampled points are within six-pixel distance from the tumor mask built by step 1. The second model has a single-size patch input, which is 13 by 13, and we combine classes 1 and 4 into a bigger class which helps us to better determine the regions of tumor core.

The third and fourth models jump from 2D models to 3D ones, and adopts the FCNN setting instead of labeling only the center of the corresponding input patch. For the third model, we choose the patch size to be $25 \times 25 \times 25$, with 8 convolutional layers of kernel size being $3 \times 3 \times 3$, and 3 convolutional layers with $1 \times 1 \times 1$ kernel size after that before the final softmax layer which produces the probability map in a $9 \times 9 \times 9$ region. This model enables us to get the labeling of the $9 \times 9 \times 9$ pixels around the center of the input $25 \times 25 \times 25$ cubic other than predicting only the center of input patch as the normal CNN model. The numbers of filters for the first 8 convolutional layers are 30, 30, 40, 40, 40, 50, 50, 50 and for the latter 3 convolutional layers are 400, 200, 150. We pick $3 \times 3 \times 3$ because replacing one convolutional layer with $5 \times 5 \times 5$ kernel by two layers with $3 \times 3 \times 3$ kernel can do the same dimension reduction but having much fewer parameters to estimate during the training process to the model with $5 \times 5 \times 5$ ones, which meets our aim of using simple but efficient model design.

The fourth model still uses the FCNN architecture but the size of input patch becomes $13 \times 13 \times 13$ and the output size is reduced to $3 \times 3 \times 3$, with five convolutional layers of $5 \times 5 \times 5$ kernel. The smaller input and output sizes may help catch more local features compared to the third model.

To combine the results of the four models used in phase2, we extract the probability of the four classes non-tumor, necrosis core, edema, enhancing core at each pixel predicted by each individual method as the candidate 'voters' and then use XGBoost to train a classification model and decide the important probabilities by cross-validation to refine the pixel-level classification. Specifically, all the sample points used by XGBoost here are selected from those pixels which have different prediction labels by the four CNN (FCNN) models in phase 2. Our empirical results shows that XGBoost could do a better job in combing the prediction results of the four models than the majority voting or averaging their probabilities. The framework diagram of our method is depicted in Fig. 2 and illustrated in Fig. 3 by one example, which shows how the initial segmentation is improved after each segmentation step.

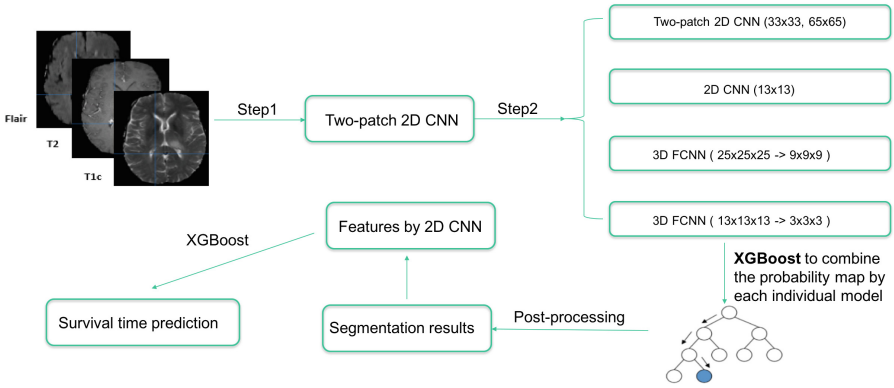

Fig. 2. Framework of TPCNN procedure

3.3 Post-processing

After using XGBoost to refine the results by combing several models in phase 2, we further improve the segmentation results by employing the following post-processing techniques. The idea is to fill in the holes and delete the small isolated clusters which are proved useful by [15]:

1. Segment the tumor mask into all connected components/clusters. Voxels in clusters whose volume is less than 0.2 times the largest connected cluster volume will be reclassified as non-tumor.
2. Segment the enhancing core mask into all connected components/clusters. Voxels in clusters whose volume is less than 0.01 times the largest connected cluster volume will be reclassified as the necrosis.

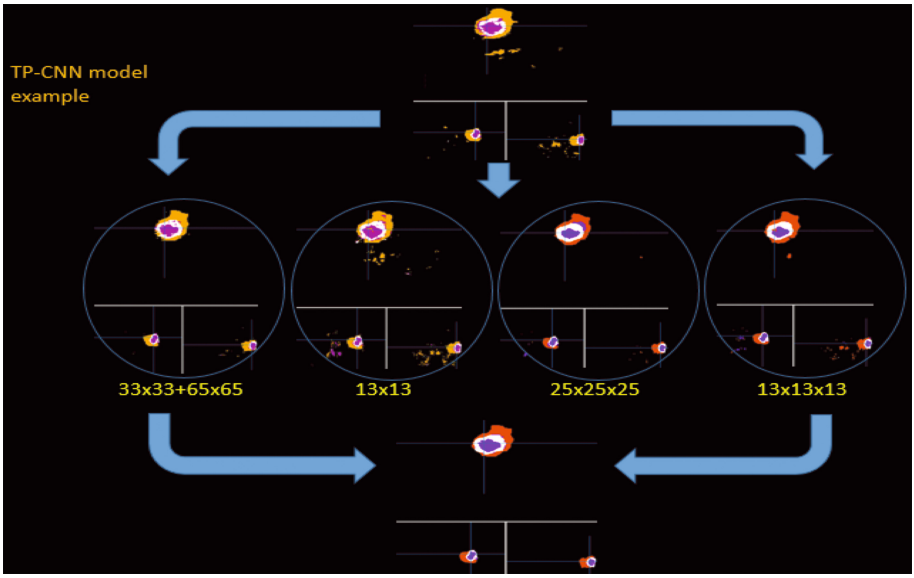

Fig. 3. Example segmentation through TPCNN procedure

3. Fill in the holes within the tumor mask and assign voxels within the holes to necrosis area.

By comparing the dice-ratios of the defined tumor sub-regions by our TPCNN model with and without the three-step post-processing, we find that the performance could be improved by applying the post-processing on the existing TPCNN results after the two-phase prediction.

4 CNN-Based Survival Time Prediction

The basic idea of our procedure to do the overall survival prediction with the brain scans is to extract the important features by training a CNN model whose input is a selected axial slice with biggest tumor regions for each subject. Continuous survival time of training subjects is divided into three categories ($<15, 10-15$ and >15) for the CNN model training. After the training is done, the last fully-connected layer before the softmax layer is considered as important features, which will be incorporated with age and some other quantitive features to predict the survival time by XGBoost regression model.

XGBoost is short for Extreme 'Gradient Boosting', where the term 'Gradient Boosting' is proposed in the paper Greedy Function Approximation: A Gradient Boosting Machine, by Friedman [2]. In brief, XGBoost is a tree-based boosting algorithm, which performs well in supervised learning problems and previous challenges. The advantage of using XGBoost here to predict the survival time

for each patient is that we could throw out the unimportant features by based on AUC value and avoid over-fitting.

We extract various quantitative imaging features directly from the tumor regions of the brain scans by simply measuring the shape, volume or other potentially important parameters as predictors to add in the survival time prediction model. Among these features, nine are describing the shape of the tumor regions, including the volume of the whole tumor, superficial area of the whole tumor, roundness and ratio between different subregions. In the training data set, all these features are computed based on the provided true annotations, but only using the segmentation results by our TPCNN model discussed in last section for the subjects in validation set. We also consider other 10 intensity statistics features (such as maximum, minimum, median, kurtosis), 57 texture features(13 gray-level co-occurrence matrix features, 8 statistics of local binary patterns and 36 threshold adjacency statistics) and 72 wavelet transform features, all of which are computed according to the tumor subregions from the corresponding modality.

Besides the global feature mentioned above, we also train a 2D convolutional neural network and extract the last dense layer as supplementary features. We take five successive 2-D slices in X-Y space with the largest number pixels belonging to class 1 and 4 for each of T1Gd, T2 and Flair scan as the three input channels. Considering the small sample size we have for the survival time prediction task (163 subjects in the training set with survival time information), we augment the sample by flipping, rotating and doing pixel shifting to the original five slides for each subject. According to the evaluation scheme based on classification principle, the continuous survival time will be divided into three groups, i.e. long-survivors (e.g., $\geq$15 months), short-survivors (e.g., $\leq$10 months), and mid-survivors (e.g. between 10 and 15 months). So we do the same way to divide the survival time for all the subjects into three classes, which are used as output to train a 2D-CNN model by borrowing the VGG-16 setting [11] and extract the dense layer before the softmax layer as the hidden image features, which will be combined with the global features and age for each subject as predictors to build the final prediction model (Fig. 4).

We also try to move to 3D CNN model instead of the current 2D to extract the feature within the tumor regions roughly masked out by segmentation results because the way we use to select 2-D slides depend more on the accuracy of segmentation predictions on validation data. However, according to the results of empirical study, the features by 3D model does not work better than those by 2D model, so we keep the 2D setting which may also help reduce a lot of computational complexity.

Before building the XGBoost prediction model, we first use Pearson chi-square test to analyze the marginal correlation between all the candidate predictors with the survival time, and sort them by p-value from small to large, where the covariate with smaller p-value may show more significant association with the survival time, and we then use the XGBoost to gradually add in covariates until the mean squared error value will not decrease after a certain

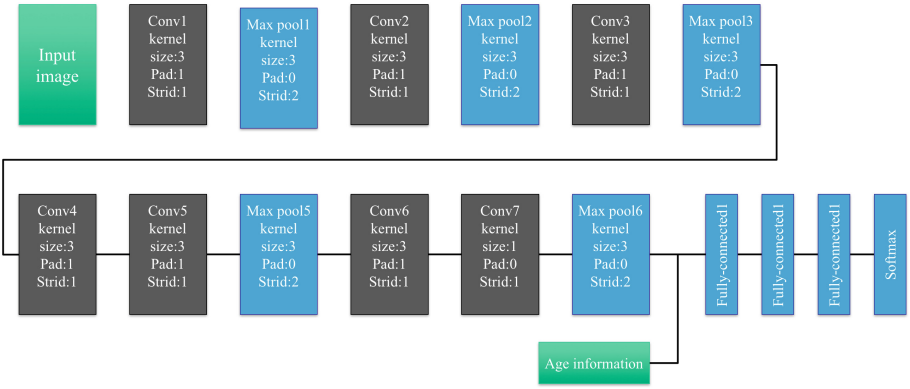

Fig. 4. 2D CNN architecture to extract important image features for survival time prediction

number of steps. To test the performance of our model, we still randomly split the training data into two parts, one of which is used to train the model and the other used to evaluate the prediction accuracy. We do 10-fold cross-validation to test the performance of our method and use the classification evaluation by dividing the survival time into three classes using threshold 10 and 15 to compute the prediction accuracy. According to the results of cross-validation, the whole tumor volume, the tumor core (necrosis core plus enhancing core) volume, the enhancing tumor volume, the longest diameter along the X-Y and X-Z space, supplementary 120 latent features achieved by CNN model are considered important and selected out, which are used to train the final model with all the training subjects to predict the survival time of subjects in validation and testing data set.

5 Empirical Study

We use the BraTS 2017 data set to test the performance of our method. We include all the 285 released subjects in the training data set to generate the training sample for phase 1 and phase 2. In particular, we split the whole training set into a sub-training set which is used to train the two-patch CNN model in the first phase and the four 2D or 3D CNN (FCNN) models in phase 2, and a sub-testing set used to validate the performance of the trained model. Specifically, the sub-training set contains 200 subjects while the other 85 subjects make up the sub-testing set. To guarantee the validation of the sub-testing data, we include 50 LGG subjects in the sub-training set and the other 25 in the sub-testing set to make sure that the proportions of LGG and HGG subjects are roughly the same in both sides.

In phase 1, we pick out 100 subjects from all the five groups, and within each of the four classes randomly sample 500 pixels from each of the four classes as

the centers to generate the 33×33 and 65×65 patches around them for each of the three modalities T2, T1Gd and Flair, where the two sizes are determined by our experiment when comparing with other sizes. The total sample size will be around $2 \times 100 \times 4 \times 1000$ where non-tumor and background pixels are combined into one class. There are many more pixels belonging to the non-tumor class in brain regions, but we sample equal number for the four classes to avoid the unbalanced bias. We do not use T1 here since the prediction result is not significantly improved when it was added in, which may occupy more computation resources and increase the running time. Besides the two kinds of input patches with different sizes as the three-channel input, we also record the label of the center voxel for each patch as the output to train the model.

At each training epoch, a small sample consists of 200 randomly selected patches (100 for each size) is used to update the model parameters. The learning rate is set to be 10^{-3} at first and then reduced to 10^{-4} after several epochs. The training process is terminated until the overall prediction accuracy of the sub-testing set when applied the updated model does not change within 20 successive training epochs.

In phase 2, within the mask built by the two-branch CNN model in phase 1, we sample the same number of patches as phase one for the first two models, while for the third and fourth 3D FCNN model, we only sample 100 patches for each class and each subject because of the memory limitation of the machines since the 3D patches takes much more space than a 2D patch. As comparison with the performance achieved by phase-one model only, we still use the same sub-training and sub-testing sets divided before the first stage. For all these four CNN (FCNN) models with different settings, we still initialize the learning rate as 10^{-3}, which will be reduced to 10^{-4} after several epochs and choose the batch size as 100. Early stopping is also applied when the performance on the sub-testing set does not improve within 20 epochs.

After training and validating the model by using the 285 training subjects, we apply the saved optimal model on the validation data which is separated from the whole training set to test the transferring ability of our model. Different from the training subjects, all the 46 subjects in the validation set do not provide the information that whether they are classified as LGG or HGG. More HGG or LGG proportions of the validation set compared to the training set may result in the lower prediction accuracy since the important features may be changed. The performance by phase-one CNN only, phase-two CNN only and our TPCNN method is shown in Table 1 in terms of the dice-ratio of each tumor regions, corresponding sensitivities and 'Hausdorff distance' statistic.

All the training and predictions are done by using a GPU sever with Nvidia K80 cards, 2496 CUDA cores, and 12 GB RAMs. The training of Phase 1 takes around 6–7 h when taking around 500 epochs until the loss value does not change within 20 successive steps. The four models in phase 2 will only take 2–3 h before we terminate the training, which costs much less time compared with the models with more complicated structures.

Table 1. Results comparison among step1, step2 and final combined two-phase results

Method	Dice_ET	Dice_WT	Dice_TC	Sensitivity_ET	Sensitivity_WT	Sensitivity_TC
Phase1	0.472	0.630	0.526	0.722	0.921	0.668
Phase2 Model1	0.594	0.832	0.670	0.674	0.911	0.759
Phase2 Model2	0.632	0.843	0.654	0.669	0.813	0.716
Phase2 Model3	0.670	0.856	0.670	0.767	0.856	0.719
Phase2 Model4	0.620	0.818	0.658	0.787	0.915	0.743
TP-CNN XGBoost	0.730	0.880	0.722	0.783	0.916	0.789

For the task of survival time prediction. We do 10-fold cross validation with the training data to test the performance of our approach and the highest prediction among the 10-times could reach 85% and the mean accuracy is around 74% (while the lowest accuracy could be around 60%). The average MSE (mean square error) of the 10-times experiments is around 89572.0. Then we use all the 160 subjects from the training sample to train the model and make predictions for the 33 validation subjects based on our segmentation results and the results are shown in Table 2.

Table 2. Evaluation of the survival prediction model by cross-validation

Survival Prediction	Accuracy	MSE	MedianSE	stdSE	SpearmanR
	0.636	216572.0	44742.6	475483.1	0.38

6 Conclusion and Discussion

Our two-phase convolutional neural network together with the XGBoost and postprocessing refinement is an automatic brain tumor segmentation method using deep learning. According to the performance of our method on the validation data set, our method could make an acceptable segmentation prediction for the non-tumor and three tumor regions in terms of dice-ratio and Hausdorff distance. One advantage is that we could achieve a high sensitivity, which means that we do not have much tumor regions to be considered as non-tumor part. Another important characteristic of our method is that we adopt some models with complicated structures, which may require a lot of computation power, but to combine the results from several simple models by using XGBoost, which is much cheaper in time consuming and machine cost. For the survival time prediction task, we use CNN model to extract the features from each of the four modalities and then use XGBoost to build the regression model, which could provide a relatively good prediction accuracy considering the small sample size we have.

We have to admit that there are several obvious weaknesses of our method. First, the pixel-based prediction procedure may result in some singular points

with a misclassified label within a regions having correct predictions, which may cause a huge reduction in the tumor core dice-ratio. Second, compared with U-Net [8] or other slice-based method, our TPCNN model could not acquire a smooth segmentation results, even though it can be improved by doing post-processing. Third, we only use simple post-processing procedures to refine the segmentation acquired by the two-phase CNN model, while some other more complicated techniques such as Conditional Random Filed (CRF) [12] are proved to greatly improve the results. Fourth, with the limitation in the machine memory, we do not sample enough patches for each individual subject compared to the total number of pixels from a corresponding modality, which may affect the power of the trained model in predicting the validation and testing data set. Despite these potential disadvantages, the simple structure and the combining idea of our method will still be valuable when there are limited computing power and time, to build a reasonable results in segmenting tumor regions and predicting survival time of patients.

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